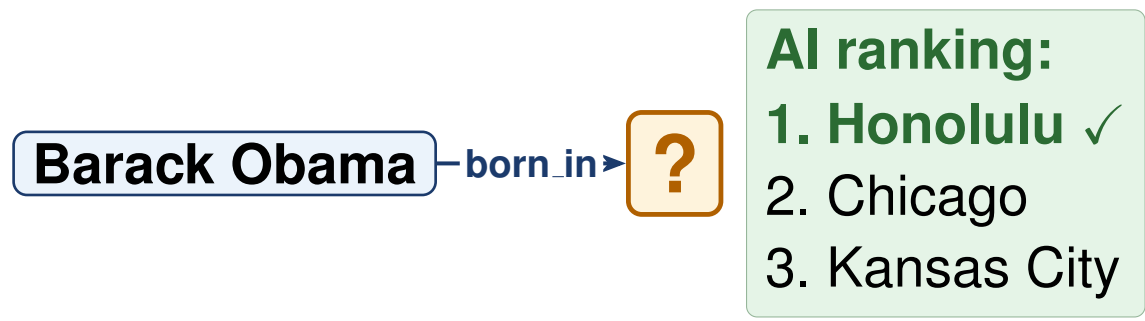


Problem

A **knowledge graph** (KG) is a network of facts stored as triples (**subject, relation, object**). **Link prediction** asks: given $(h, r, ?)$, which entity t is most likely to complete the triple?



Models learn by comparing correct triples against *negatives* (fabricated wrong triples). The quality of negatives is critical.

RANDOM NEGATIVES (STANDARD)

Obama → born.in → Ice cream

Obama → born.in → Mt. Everest

Obama → born.in → Pacific Ocean

Too easy. Model stops improving.

HARD NEGATIVES (OURS)

Obama → born.in → Chicago

Obama → born.in → Kansas City

Obama → born.in → Honolulu

Plausible but wrong. Model keeps improving.

Honolulu is the correct answer here, making it the hardest possible negative for this query.

Motivation

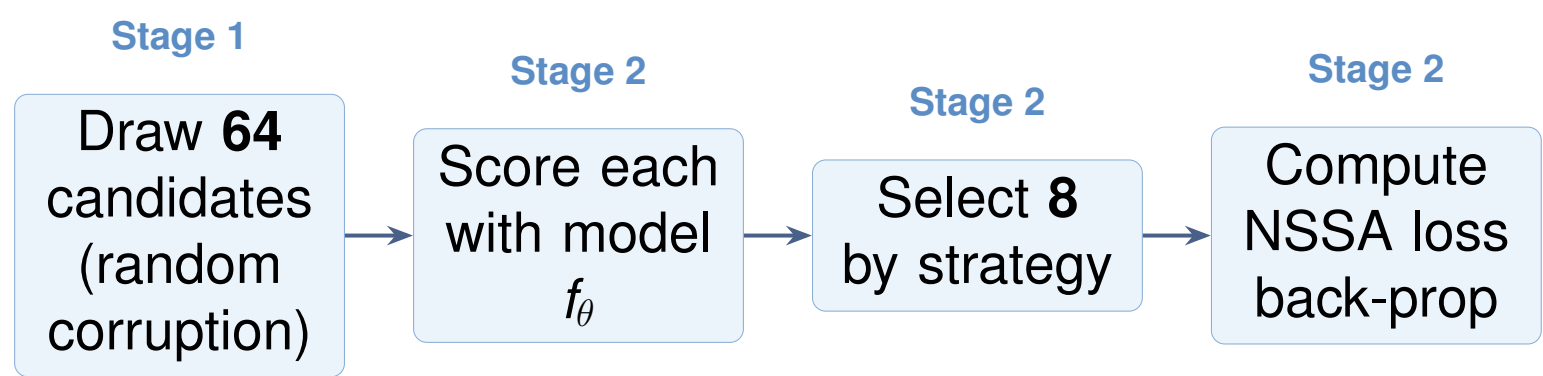
Random negatives (gray box above) become trivial after a few epochs: the model rejects them easily, so the **gradient signal vanishes** and learning plateaus.

Hard negatives (amber box) keep the model challenged, but training only on them can destabilise learning.

Our research question. Can a **mixture** of random and hard negatives combine the accuracy of hard with the stability of random — and what is the best proportion?

Methodology

For each positive triple at training time:



Selection strategies tested in Stage 2:

- **Random:** sample 8 uniformly (control)
- **Hard:** take the 8 highest-scoring wrong candidates
- **Mixed p :** take fraction p hard + $(1 - p)$ random, with $p \in \{30\%, 50\%, 70\%\}$

All strategies share the same Stage 1 pool: zero extra compute.

Common setup: RotatE, embedding dim 128, batch 1024, Adam lr= 10^{-3} , 50 epochs (patience 10), seed 42.

References

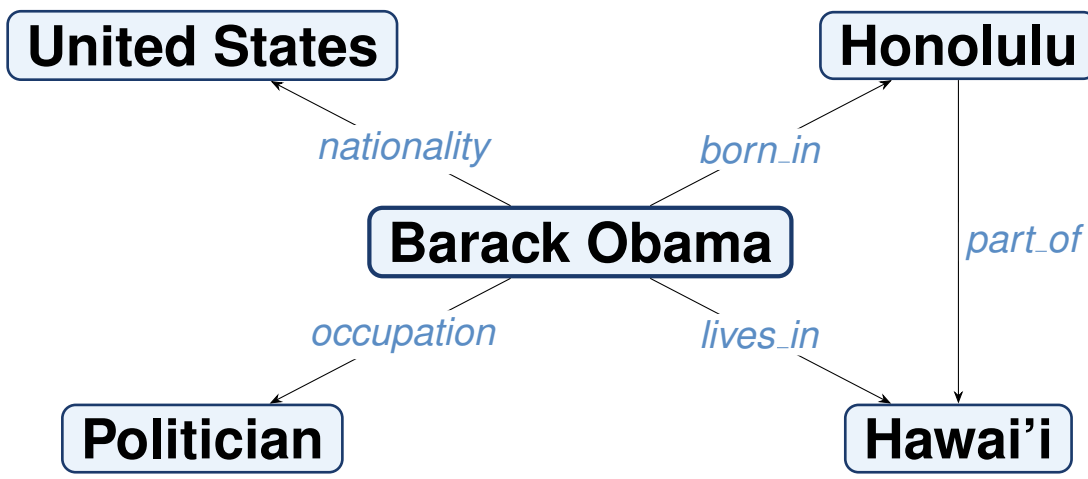
[1] Sun *et al.* (2019). *RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space*. ICLR 2019.

[2] Cai & Wang (2018). *KBGAN: Adversarial Learning for Knowledge Graph Embeddings*. NAACL 2018.

[3] Toutanova & Chen (2015). *Observed versus latent features for KB and text inference*. ACL Workshop 2015 (FB15k-237 dataset).

Datasets

FB15k-237 is a widely used benchmark knowledge graph derived from Freebase, covering people, places, films, and organisations. Each node is an **entity**; each labelled edge is a **fact**.



Entities: 14,505 Train triples: 272,115
Relation types: 237 Val / Test: 17,526 / 20,438
Relation frequencies range over **three orders of magnitude** (37 to 15,989 occurrences), enabling stratified evaluation by rarity.

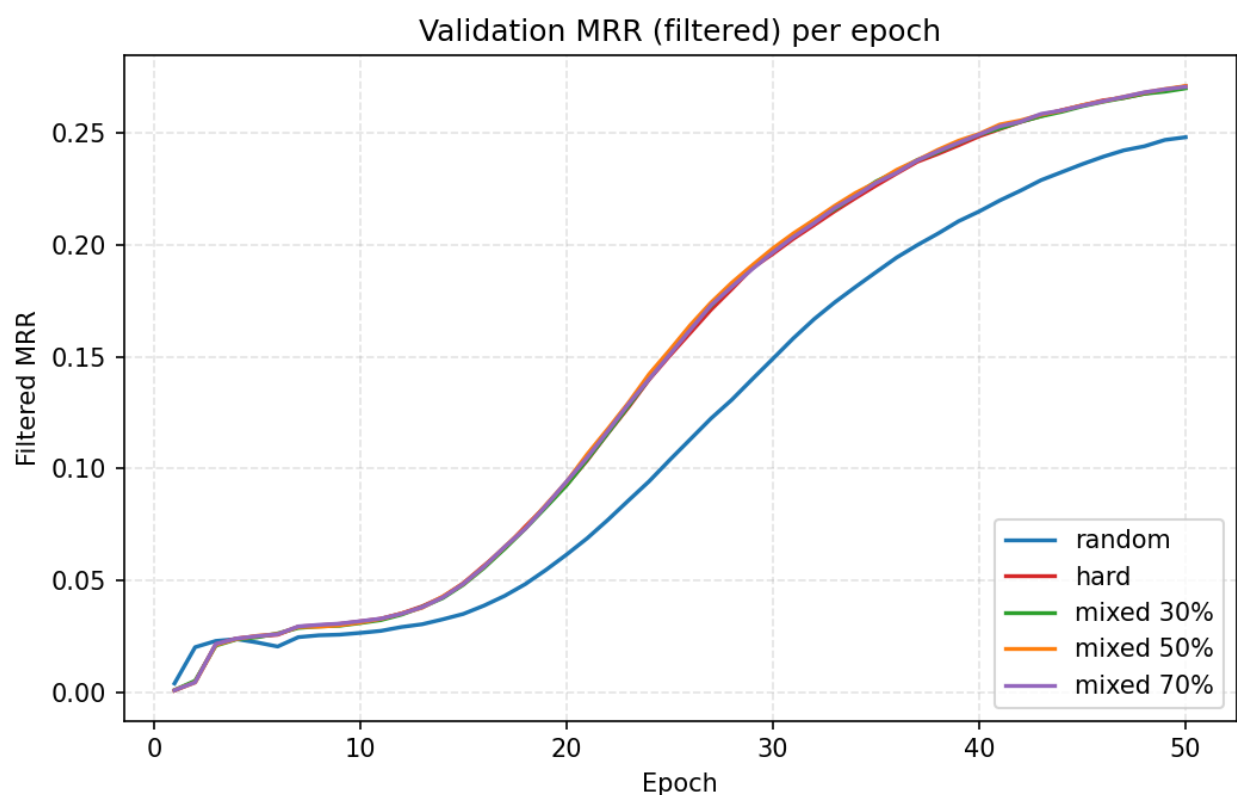
Results

+10.4%

accuracy gain (MRR) with Hard negatives vs. standard random sampling

Strategy	MRR	H@1	H@3	H@10
Random (baseline)	0.270	0.189	0.300	0.428
Hard	0.298	0.217	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	0.328	0.460
Mixed 70%	0.299	0.217	0.328	0.461
Δ Hard – Random	+0.028	+0.028	+0.026	+0.031

Mixed 30% hard recovers 95% of the gain at zero extra cost.



Hard and mixed strategies **pull ahead from random after epoch 15** and converge to a 7–8 % higher plateau with no training instability.

Stratified evaluation, MRR gain Hard vs. Random:

+3.6 pp

Rare relations

0.382 → 0.418

+3.0 pp

Medium relations

0.332 → 0.362

+2.8 pp

Frequent relations

0.257 → 0.285

Hard negatives benefit every slice, but gain is largest for **rare relations** where random negatives are least informative.

Conclusions

- ▶ **Hard negatives are a free upgrade:** same training pipeline, smarter selection step, +10.4% MRR gain.
- ▶ **You don't need 100% hard:** Mixed 30% (2 hard examples out of 8) recovers 95% of the gain with added stability.
- ▶ **Rare and ambiguous facts benefit most**, which is exactly where KG completion is hardest and most valuable in practice.
- ▶ **Future work:** dynamically adjust the hard fraction during training; scale to Wikidata-5M; test on RESCAL and TuckER.